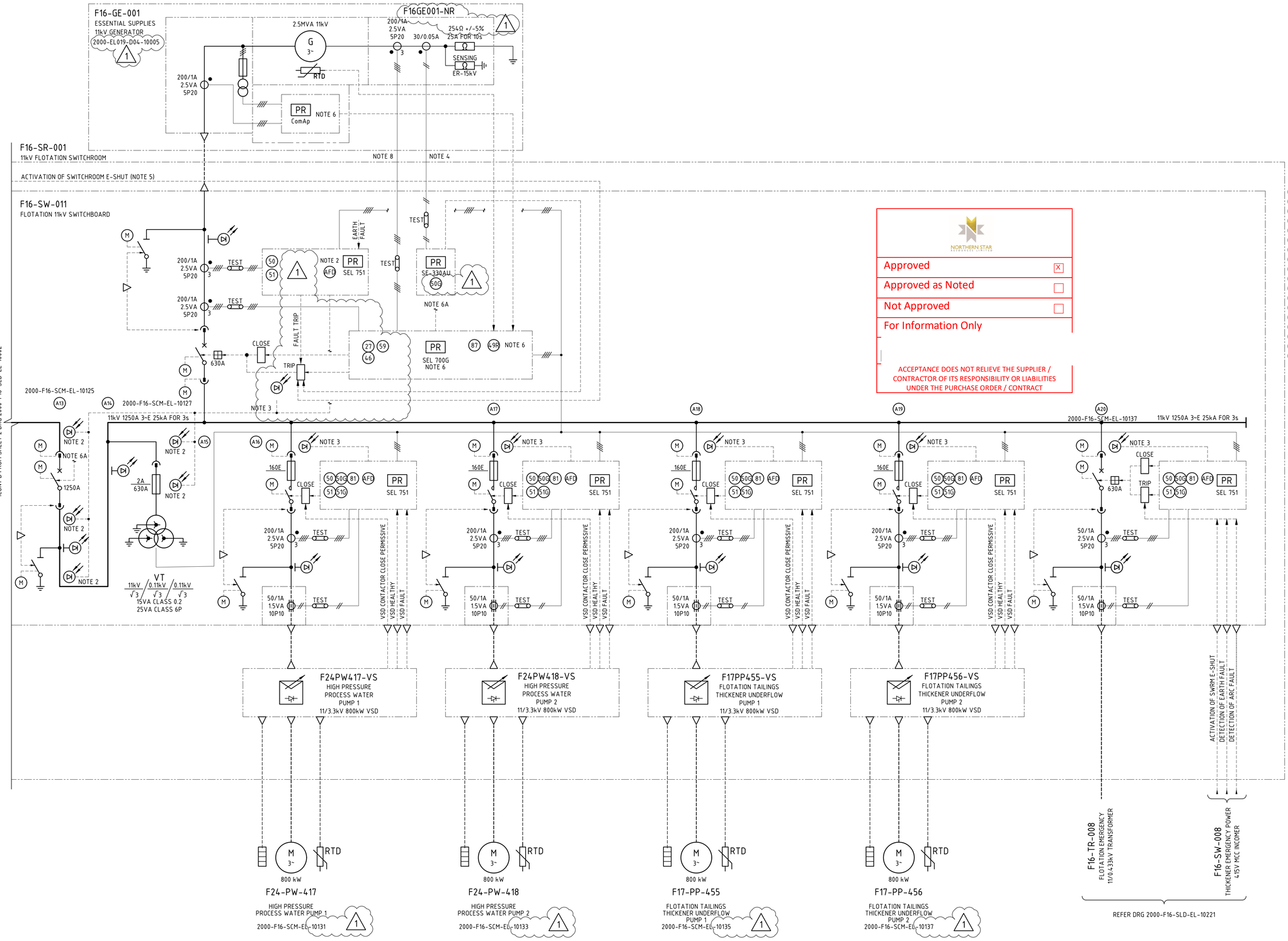


NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION .
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER.
6. NORMAL OPERATING CONDITIONS :
   A) DCS MONITORS THE MAIN BUS VOLTAGE: UPON DETECTION OF MAINS LOSS, DCS TO INITIAL TRANSFER SEQUENCE.
   B) DCS TO OPEN BUSTIE CB, START GENERATOR AND CLOSE GENERATOR INCOMER CB WHICH ENERGISES THE RHS 11kV BUS.
   C) DCS MONITORS THE EMERGENCY BUS VOLTAGE TO CONFIRM POWER IS AVAILABLE.

   REINSTATEMENT - MAINS POWER RESTORED :
   A) DCS MONITORS THE MAIN BUS VOLTAGE: UPON DETECTION OF MAINS IS RESTORED, DCS TO PROMPT THE OPERATOR.
   B) OPERATOR TO INITATE TRANSFER SEQUENCE ON DCS.
   C) DCS TO STOP GENERATOR, OPEN GENERATOR INCOMER CB, CLOSE BUSTIE CB WHICH ENERGISES THE RHS 11kV BUS.
   D) DCS MONITORS THE EMERGENCY BUS VOLTAGE TO CONFIRM POWER IS AVAILBLE.
7. GENERATOR ENGINE START AND CONTROL CIRCUIT SHALL BE KEY INTERLOCKED WITH EMERGENCY BUS INCOMER CIRCUIT BREAKER EARTH SWITCH TO PREVENT STARTING OF GENERATOR WHILST EMERGENCY BUS INCOMER CIRCUIT BREAKER EARTH SWITCH IS CLOSED.
8. ONE SET OF PROTECTION CT's SHALL BE INSTALLED ON THE ALTERNATOR SIDE OF THE 11kV GENERATOR FOR WINDING PROTECTION.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
25	SYNCHRONISATION CHECK
27	UNDER VOLTAGE
46	REVERSE PHASE OR PHASE BALANCE CURRENT OR STATOR CURRENT UNBALANCE
49R	RTD BIASED THERMAL OVERLOAD
50	INSTANTANEOUS OVER-CURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
74 TCS	ALARM - TRIP CIRCUIT SUPERVISION
81	FREQUENCY
81O	OVER FREQUENCY
81U	UNDER FREQUENCY
86	LOCKING OUT
87	DIFFERENTIAL PROTECTION
AFD	ARC FLASH DETECTION

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Single line diagram sheet 3